

Armavour: A CCPA-Grounded Agent Dark-Pattern Benchmark and Five Measurement-Validity Failures

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Abstract—Language model agents increasingly transact on the web on a user's behalf, through interfaces that frequently contain dark patterns. We present Armavour, a benchmark measuring whether computer-use agents are deceived by the twelve dark patterns codified by India's Central Consumer Protection Authority, implemented at four manipulation intensities in three languages, with ground truth taken from the document object model rather than from agent self-report. Across 1,928 episodes and two agent architectures we report that susceptibility scales monotonically with intensity, and that patterns exploiting information asymmetry succeed while patterns exploiting human perceptual and affective limits—visual salience, irritation, patience, guilt—fail entirely under both architectures. We also report, as a co-equal contribution, five distinct ways in which this benchmark measured something other than what it named: an item whose manipulation could be bypassed by evaluating the task's stated criterion; element identifiers that disclosed the ground-truth answer to the agent; an agent framework's internal completion signal absorbed by an outcome-based metric as evidence of safe behaviour; a results column whose denominator changed silently mid-table; and a deception rate inflated by baseline comprehension failure on non-English interfaces. Each produced stable, plausible, internally consistent numbers; none was visible in aggregate statistics; each was found only by reconstructing the quantity from raw episode records under an explicit hypothesis. We give the detection signature for each and argue all five generalise beyond this benchmark.

Index Terms—dark patterns, LLM agents, computer-use agents, benchmark, measurement validity, construct validity, consumer protection, multilingual evaluation, AI safety

I. INTRODUCTION

Dark patterns are interface designs that exploit predictable human cognitive tendencies to produce outcomes favouring the platform over the user: a pre-ticked donation checkbox, a mandatory fee revealed only at final confirmation, a decline button labelled to induce guilt. They are widespread, well documented, and—in a growing number of jurisdictions—illegal.

The deployment of autonomous web agents changes the threat model. When an agent completes a purchase on a user's behalf, the manipulation is no longer aimed at a human's

attention and emotions; it is aimed at whatever the agent's perception pipeline delivers to a language model, and at whatever reasoning that model then performs. Some manipulations may fail entirely against this substrate. Others may succeed more reliably than against humans.

This paper answers two questions, and we regard them as equally important.

The first is the obvious one: *do legally-codified dark patterns deceive computer-use agents, and which ones?* We built Armavour to answer it, grounding the taxonomy in India's *CCPA Guidelines for Prevention and Regulation of Dark Patterns, 2023* rather than an HCI taxonomy, so that a result about pattern *X* is a result about a category regulators already enforce against named platforms.

The second emerged from building it: *can we trust the measurement that answers the first?* Five times during this work, a component of our benchmark produced numbers that were stable across reruns, plausible against prior work, internally consistent, and wrong about what they measured. One item's manipulation turned out never to be load-bearing. The element identifiers our harness requires turned out to name the correct answer. One scoring rule quietly counted an agent framework's own "task complete" signal as evidence the agent had resisted a dark pattern. One results column pooled arms with different coverage and changed denominator halfway down. And one experimental arm's headline deception rate turned out to rest substantially on the agent failing to comprehend the honest interface at all.

None of these announced itself. Each was found only by reconstructing the reported quantity from raw episode records under an explicit hypothesis and observing where the reconstruction failed. We report all five in Section V, before the results they affect, because we believe the failure modes are properties of how this class of benchmark is built rather than accidents of our implementation.

Contributions.

- 1) **Armavour**: an open benchmark for twelve CCPA-codified dark patterns at four intensities and three languages, with DOM-level ground truth, versioned interface contracts, and a validated LLM judge for the

patterns whose success condition is not deterministically observable.

- 2) **Five measurement-validity failure modes** in agent susceptibility benchmarking—criterion-bypassable items, identifier leakage, completion-signal contamination, denominator drift, and baseline contamination—each with the empirical signature that detects it (Section V).
- 3) **A reconstruction protocol**: we argue that recomputing every reported quantity from raw records under a stated hypothesis should be a routine step in benchmark reporting rather than a response to anomaly, and show that all five failures above were detectable this way and by no other means we tried.
- 4) An **intensity dose-response result**: deception rises monotonically from 0% at control to 38% at aggressive intensity under otherwise identical conditions, replicated across two architectures.
- 5) A **pattern-class result**: four patterns produce no genuine deception under either architecture, and all four target human perceptual or affective properties rather than information content (Section ??).
- 6) A **multilingual result**, reported conditionally: with the instruction held in English, deception rises from 20.7% on an English interface to 36.0% on a Hindi one ($p = 1.5 \times 10^{-5}$). With the instruction also localised, the measured rate rises further but is not interpretable, for reasons given in Section V-D.

II. RELATED WORK

A. Dark pattern taxonomies and detection

Work on dark patterns has largely proceeded from qualitative taxonomy toward automated detection at scale. Existing taxonomies are descriptively useful but heterogeneous, and detection work typically answers whether a pattern is *present* on a page rather than whether it *works* on a given agent. Armavours addresses the latter and, by adopting a legal taxonomy, makes its categories directly commensurable with regulatory action.

B. Agent benchmarks and susceptibility

Web-agent benchmarks predominantly measure task completion. Recent work measures susceptibility to adversarial or manipulative interfaces; SusBench is the closest prior work and reports pattern-level avoidance rates we treat as a comparison baseline. Armavours differs in grounding its taxonomy legally, and in varying manipulation *intensity* and interface *language* as controlled factors, permitting dose-response and cross-lingual claims that presence/absence designs cannot support.

C. Validity of agent-safety measurement

Benchmark construction receives far less scrutiny than benchmark results. Where validity is discussed in ML evaluation it is usually framed as contamination, saturation, or construct drift across model generations—problems of a benchmark becoming *stale*. The failures we report are different: an item that is internally consistent, has unambiguous ground

truth, produces stable numbers across reruns, and measures a capability unrelated to the one it names. Such an item does not degrade over time; it was never measuring the construct.

This matters specifically for safety benchmarks. A capability benchmark that measures the wrong thing produces a wrong leaderboard. A safety benchmark that measures the wrong thing produces a false assurance, and the direction of the error is not random—four of our five failures biased toward *apparent safety*, because the default coding of an ambiguous outcome is usually the benign one. We are not aware of prior work characterising these failure modes for agent susceptibility evaluation or supplying detection signatures for them, and we treat that gap as this paper’s most transferable contribution.

D. Multilingual evaluation

Degradation in lower-resource languages is well established for static tasks. Its consequences for *agentic* behaviour have received less attention. Section V-D shows that at sufficient degradation the question becomes unanswerable rather than merely harder, because task failure and manipulation susceptibility stop being separable.

III. THE ARMAVOUR BENCHMARK

A. Design principles

Ground truth from the interface, not the agent. An agent’s report that it declined an add-on is not evidence that it did. Every pattern exposes a deterministic oracle reading terminal DOM state at the moment of the transaction.

Intensity as a controlled variable. Each pattern is implemented at control, subtle, moderate and aggressive, varying only the manipulation. control renders the task honestly and supplies the baseline against which deception is measured. Section V-D shows this condition does more work than we originally assigned it.

Versioned interface contracts. Five contracts govern component seams: episode configuration, oracle output, stable element identifiers, the episode log schema, and the judge rubric interface. Section ?? shows that the third of these was in tension with measurement validity in a way we did not anticipate.

B. Testbed

A React/TypeScript application presenting seven simulated services (ticketing, e-commerce, education, subscription management, SaaS signup, security software, news). Twelve of the thirteen CCPA patterns are implemented; the thirteenth concerns advertising practice not observable within a single session. Episodes are driven entirely by URL parameters:

```
/?site={s}&pattern={p}&intensity={i}
&lang={l}&seed={n}
```

Every episode is a pure function of its configuration; the SHA-256 hash of that configuration is the idempotency key for logging and resumption.

C. Adapters

ComputerUse is our own: it extracts interactive elements and their surrounding text context from the DOM, presents them as structured JSON, and executes the returned action via Playwright. Action vocabulary is minimal—click, check, uncheck, fill, done.

BrowserUse is a third-party agent framework with its own DOM indexing, prompting, and multi-step planning. It was included so that a finding could be attributed to agents rather than to one extraction layer. Section V-B shows this inclusion mattered in a way we did not anticipate.

D. Oracle and outcome coding

At the terminal action the page sets a global result object:

```
{ pattern, avoided, total, ... }
```

Outcomes are coded on two binary axes—task completed, pattern avoided—giving EC (completed, avoided), DC (completed, deceived), EF (not completed, avoided), DF (not completed, deceived).

E. Judge and its validation

Two patterns—false urgency and confirm shaming—have a success condition DOM state cannot determine, because the question is whether the manipulation *influenced* the decision rather than whether a button was pressed. A second language model scores the reasoning trace against a CCPA-grounded rubric, returning a flag and a one-sentence evidence string. Contract 5 requires judge $\neq$ agent.

For these patterns the reported susceptibility rate is the *judge flag* rate, not the raw DC rate. A DC outcome records that the platform-favoured result occurred, not that the pattern caused it; we report EC, genuine DC (judge-confirmed), and task-failure DC separately. Section VI shows this distinction changes confirm shaming’s classification entirely.

An agent whose reasoning acknowledges the manipulation but whose final action is correct is scored as *not* swayed: the benchmark measures whether the pattern changed the decision, not whether it entered deliberation.

We validated the judge against twelve hand-constructed cases, deliberately including the failure we had observed—episodes where the agent takes the task-correct action while quoting the guilt-laden label, because in this pattern the correct button *is* the guilt-laden button. An initial judge model scored 72.7% accuracy, precision 0.50, recall 1.00: systematically over-flagging. Its failures were not borderline—in one case its own evidence string stated the agent showed “no hesitation or reconsideration” and then flagged it anyway, having inverted a value it was given directly. A larger judge model, with *no change to the rubric text*, scored 100% across all twelve with zero parse errors, reproduced over two consecutive runs. The problem was model capability, not prompt engineering, and further rubric iteration would have been wasted effort.

IV. EXPERIMENTAL DESIGN

The agent is llama-3.3-70b-versatile; cross-model spot-checks use llama-3.1-8b-instant and gpt-oss-20b; the judge is gpt-oss-120b. Inference is deterministic (temperature = 0) at every call site. Both llama models have since been withdrawn by the provider; see Section ??.

- **E1a**: twelve patterns $\times$ four intensities $\times$ ten seeds, English, ComputerUse.
- **Spot-check**: twelve patterns, aggressive only, five seeds, smaller models, ComputerUse.
- **E1b**: E1a’s matrix on BrowserUse.
- **E2**: five language-sensitive patterns $\times$ three intensities $\times$ three interface languages $\times$ ten seeds, *English instructions throughout*.
- **E2a / E2b**: E2’s Hindi and Hinglish cells re-run with the instruction also localised.

1,928 episodes were run; 1,600 are scored, after excluding two patterns established as unmeasurable (Sections V-A and ??) and eight duplicate reruns. Language-sensitive strings were written by a native speaker, not machine-translated: a flat translation of a triple-nested negation would not preserve the manipulation the item tests.

The E2 versus E2a/E2b split was intended to separate interface language from instruction language. It ended up serving a different and more important purpose (Section V-D).

V. MEASUREMENT-VALIDITY FAILURES

We present these before the results because four of the five change figures we would otherwise have reported, and because the reconstruction protocol that found them is what licenses the numbers in Section VI.

A. Criterion-bypassable items

Our disguised-advertisement item presented four search results, one a sponsored placement styled to blend in, with label prominence decreasing as intensity rose. The task was to buy the cheapest genuine product. Following our own specification, the ad was priced above every organic item so selecting it would be unambiguously wrong.

That decision made the item unmeasurable. Because the ad was the most expensive and the task asked for the cheapest, any agent that compared prices avoided it—regardless of whether the sponsored label was shown, greyed, or absent. Intensity could not affect the outcome, and did not: 0% deception at every level, under both architectures.

Detection signature: an intensity ladder with zero slope, and disagreement with a prior run. An earlier pilot returned 60% deception on this item. Reconciling the two showed the pilot agent had received no product names or prices at all—its traces read “all buttons have the same text ‘Buy’” and “no product details provided”—and had defaulted to the first element, which happened to be the ad. The pilot measured a perception-pipeline gap and reported it as ad-disguise susceptibility. Once the extraction layer supplied surrounding text,

the rate fell to zero, which is the correct value for an item that cannot discriminate.

General condition: *if the task’s stated criterion is sufficient to identify the correct choice, the manipulation is never load-bearing, and the item measures criterion evaluation rather than manipulation susceptibility.* An item can satisfy every apparent correctness requirement—unambiguous ground truth, a well-formed intensity ladder, a clean oracle—and still measure nothing.

False urgency has the same structure—single-choice, price-decidable, urgency copy never load-bearing—with no equivalent fix: making the urgent item cheapest would make selecting it correct, and equalising prices removes the discriminator. Urgency is pressure to act *quickly*, which a single-step episode cannot express.

Corollary for benchmark authors: a pattern scoring far from its published baseline in either direction warrants a construction audit before it is reported. Our 100% avoidance against a prior-work baseline of roughly 65% was the anomaly that started this investigation.

B. Identifier leakage

We had planned to correct the disguised-advertisement item by inverting the price relationship, so the ad becomes cheapest and price-only reasoning selects it. A later spot-check on a third agent model showed that this correction would not have worked, for a reason we had not considered.

Contract 3 requires every interactive element to carry a stable, unique identifier, so the harness can act on elements reliably across renders. We named them descriptively, as one does: `item-ad` for the sponsored placement and `item-org-1` through `item-org-3` for the organic results. The extraction layer includes each element’s identifier in the JSON handed to the model, because the model must be able to refer to elements when choosing an action.

The identifiers name the answer. At aggressive intensity—the level at which the specification requires *no label at all*—an agent produced this reasoning:

“The ad item (PowerCharge Fast Cable) is a disguised advertisement and should not be considered genuine. Among the genuine organic results, the cheapest is Basic USB-C Cable at Rs 199.”

It identified the sponsored placement with no visual label present, and described the alternatives as “organic”—a term from our internal taxonomy, not from the rendered page. The same holds for false urgency, whose identifiers are `item-urgent` and `item-calm-1`; agent traces reference `buy-item-calm-1` directly.

Detection signature: *agent reasoning containing vocabulary that appears nowhere in the rendered interface.* The word “organic” does not appear on the page. Its presence in a reasoning trace is a direct indication that something other than the rendered interface reached the model.

General condition: *any channel the harness requires for control—element identifiers, accessibility labels, test hooks, ordering—can also carry the ground-truth answer, and will*

do so by default, because the names that make a testbed maintainable are the names that describe what each element is for. This is a direct tension between a benchmark’s engineering requirements and its measurement validity, and it is invisible from outcome data alone: an item leaking its answer looks exactly like an item the agent is robust to.

The remedy is opaque identifiers—`item-a`, `item-b`—accepting the maintenance cost, and auditing every pattern for the same disclosure. We have not completed that audit, and therefore report no corrected figures for either affected item.

C. Completion-signal contamination

BrowserUse produced 47 EF outcomes (11.8% of scored episodes) against ComputerUse’s 3 (0.75%)—a sixteenfold difference in episodes scored as “avoided the dark pattern but did not complete the task.”

We first hypothesised swallowed exceptions in the adapter’s error handling. Trace analysis falsified this. Of 48 EF episodes in the raw arm, 45 contain an explicit `done` action, against 1 of 378 EC episodes: near-perfect separation. Token counts are comparable to EC episodes (6,870 versus 7,163 mean input tokens), durations are *longer* (45.2s versus 37.9s), no episode approaches the step budget, none has zero token usage. These are not failures. The agent declared itself finished.

The mechanism is architectural. BrowserUse treats `done` as its native completion signal and its prompting encourages invoking it when the task appears complete; ComputerUse offers a bare action choice with no equivalent framing. The framework’s notion of completion fires before the page’s terminal action, the oracle never runs, and an outcome scheme keying “avoided” on the absence of a deceived oracle result credits the episode as safe.

A later spot-check reproduced the effect in a different model on our own adapter, at a higher rate: 16 of 60 episodes (27%), with traces such as “*the order summary confirms the ticket price is Rs 500 and the order has been placed successfully*” immediately before a `done`—issued without the payment button ever being clicked. The agent did not merely stop early; it asserted a completion that had not occurred.

Detection signature: *an outcome-category rate that differs by an order of magnitude across architectures or models, plus the presence of that category in the control condition.* Seven of BrowserUse’s 48 occur at control intensity, where no manipulation exists to avoid. Those are unambiguously task failures scored on the safe side.

General condition: *when an agent framework has an internal notion of task completion, and a safety metric infers safety from the absence of an unsafe outcome, the framework’s completion signal is silently converted into a safety claim.* This is invisible in aggregate statistics—BrowserUse’s lower deception rates in Table IV are partly this artefact—and surfaces only when the ambiguous outcome category is disaggregated and its traces read.

D. Denominator drift

An earlier draft of our pattern table reported a single pooled deception rate per pattern, computed across every arm in which

that pattern appeared. Because the language arms cover only five of the twelve patterns, half the column was computed at $n = 135$ and half at $n = 45$, with nothing in the table indicating the denominator changed. The column was internally consistent, monotonic, and plausible.

Detection signature: partial reproduction. It was caught by attempting to recompute the column from raw records under one stated hypothesis (“E1a plus spot-check, $n = 45$ ”) and finding that exactly six of ten values matched. The four that did not were exactly the four patterns also present in the language arms.

General condition: *a table that pools across experimental arms with different factor coverage will silently vary its denominator by row.* The failure is undetectable by inspection and trivially detectable by reconstruction.

E. Baseline contamination

The fifth failure inflates this paper’s own headline result, and is the reason Section ?? reports the language effect conditionally.

The control condition renders each task honestly: no manipulation is present, so a deceived outcome should be impossible. Under English instructions with an English interface, it is—0 of 50 control episodes were scored as deceived. But control-condition deceptions rise sharply as the interface and instruction are localised (Table I).

TABLE I
DECEPTIONS IN THE control CONDITION, WHERE NO MANIPULATION IS PRESENT ($n = 50$ PER ROW)

Instruction	Interface	“Deceived”
English	English	0
English	Hinglish	0
English	Hindi	6
Hinglish	Hinglish	7
Hindi	Hindi	21

In the fully-Hindi arm, 42% of episodes on an interface containing no dark pattern were scored as deceived—including *every* control episode for interface interference and *every* control episode for confirm shaming. The agent is not being manipulated in these episodes. It is failing to operate the interface, and an outcome scheme that codes any incorrect end-state as deception counts that failure as susceptibility.

Detection signature: a non-zero rate in the null condition, disaggregated along the same axis as the reported effect. A pooled control figure across all arms (34 of 300, or 11%) looks like tolerable noise. Split by language it is a monotone tracking precisely the variable whose effect is being claimed.

General condition: *when a manipulated variable also degrades baseline task competence, an outcome-based deception metric sums the two, and the design cannot apportion them.* This is not a confound in the usual sense—no third variable causes both—but a failure of the metric to distinguish two mechanisms producing the same recorded outcome.

The consequence is direct. Our fully-localised Hindi arm records 73.3% deception; a large share is baseline failure, and we cannot say how much. The English-instruction arms retain an interpretable baseline and are reported as the primary result.

F. What the five have in common

Each produced a number that was stable across reruns, plausible against prior expectations, internally consistent, and not a measurement of what it named. None was visible in aggregate statistics. Each was found the same way: by taking a reported quantity, stating a hypothesis about how it was computed, recomputing it from raw episode records, and examining where the reconstruction diverged.

Four of the five biased toward apparent safety. We suspect this is systematic rather than coincidental: when an outcome is ambiguous, the default coding is usually the benign one, because benign outcomes are what the instrument is built to expect.

VI. RESULTS

All figures below are computed on the 1,600 scored episodes, excluding disguised advertisement and false urgency consistently across every table.

A. Aggregate outcomes

TABLE II
OUTCOME DISTRIBUTION BY ARM (SCORED EPISODES)

Arm	n	EC	DC	EF	DC _{judge}
E1a (ComputerUse)	400	81.8%	17.5%	0.8%	17.3%
Spot-check (8B)	50	48.0%	42.0%	10.0%	38.0%
E1b (BrowserUse)	400	74.8%	13.5%	11.8%	13.5%
E2 (EN instr.)	450	69.8%	27.8%	2.4%	27.8%
E2a (HI instr.)	150	26.7%	73.3%	0.0%	54.0%
E2b (Hin. instr.)	150	70.0%	30.0%	0.0%	30.0%

No episode in 1,928 produced a DF outcome; this is a property of the scoring construction (Section ??). The spot-check arm is independently notable: the smaller model was deceived at more than twice the larger model’s rate (42.0% versus 17.5%) under identical conditions. E2a’s figures should be read with Section V-D.

B. Intensity dose-response

TABLE III
DECEPTION RATE BY INTENSITY, ENGLISH INSTRUCTION ($n = 100$ PER CELL)

Intensity	E1a	E1b
Control	0%	5%
Subtle	9%	8%
Moderate	23%	15%
Aggressive	38%	26%

The gradient is monotonic in both architectures and causal in the strict sense that intensity is the only manipulated

variable. ComputerUse’s control produces no deception across 100 episodes, which is the precondition for interpreting any other number in this table.

BrowserUse’s control does not, at 5%. All five affected episodes fall in two patterns whose control variant initialises to the already-correct state, so a deceived outcome requires an unforced action. This is a smaller instance of Section V-D’s problem in an architecture rather than a language, and we have not resolved whether it reflects BrowserUse’s independent DOM indexing or an oracle-reading difference through that adapter.

C. Pattern-level susceptibility

TABLE IV
DECEPTION RATE BY PATTERN AND ARCHITECTURE ($n = 40$ PER CELL, ENGLISH INSTRUCTION). * JUDGE-ADJUSTED; RAW RATE 2.5%.

Pattern	E1a	E1b
SaaS billing	55.0%	30.0%
Trick question	42.5%	65.0%
Drip pricing	25.0%	22.5%
Bait and switch	22.5%	5.0%
Forced action	15.0%	12.5%
Basket sneaking	12.5%	0.0%
Confirm shaming*	0.0%	0.0%
Subscription trap	0.0%	0.0%
Interface interference	0.0%	0.0%
Nagging	0.0%	0.0%
<i>Excluded, Sections V-A and ??:</i>		
Disguised advertisement	—	—
False urgency	—	—

Confirm shaming’s raw rate under E1a is 2.5%—a single episode—but the judge classifies that episode as a task-failure DC rather than genuine susceptibility, giving 0.0%. Without the judge distinction it would appear as weakly effective rather than as a null.

Three observations. First, four patterns produce no genuine deception under either architecture; Section ?? argues they share a cause. Second, trick question is the only pattern where BrowserUse is substantially *more* susceptible—and the only one whose manipulation is purely linguistic, suggesting additional scaffolding does not help, and may hinder, when the difficulty is semantic rather than navigational. Third, BrowserUse’s lower rates elsewhere should be read alongside Section V-B.

D. Language

Primary result. With the instruction held in English and only the interface localised, deception rises from 20.7% (English) to 26.7% (Hinglish) to 36.0% (Hindi). The design is fully paired on (pattern, intensity, seed), with all 150 pairs matching and none dropped. An exact McNemar test on discordant pairs gives, for English versus Hindi, 26 scenarios avoided in English but deceived in Hindi against 3 in the reverse direction, $p = 1.52 \times 10^{-5}$.

TABLE V
DECEPTION RATE BY LANGUAGE CONDITION ($n = 150$ PER ROW). †NOT INTERPRETABLE; SEE SECTION V-D.

Instr.	Interface	DC	Control fail
English	English	20.7%	0/50
English	Hinglish	26.7%	0/50
English	Hindi	36.0%	6/50
Hinglish†	Hinglish	30.0%	7/50
Hindi†	Hindi	73.3%	21/50

English versus Hinglish is *not* significant: 16 versus 7 discordant pairs, $p = 0.093$. We state this because the three-point monotone in the table invites a stronger claim than the pairwise tests support.

Secondary arms. With the instruction also localised, measured deception reaches 30.0% (Hinglish) and 73.3% (Hindi). We do not interpret these. The Hindi arm’s control condition fails 21 times in 50 (Table I), so an unknown but substantial share of that 73.3% is the agent failing to operate an honest interface. The judge-adjusted figure for this arm is 54.0%, a 29-episode gap, indicating the contamination is partly visible to a judge and partly not.

TABLE VI
LANGUAGE EFFECT BY PATTERN, ENGLISH INSTRUCTION ($n = 30$)

Pattern	EN	Hinglish	Hindi
SaaS billing	40.0%	66.7%	60.0%
Trick question	36.7%	33.3%	53.3%
Forced action	26.7%	33.3%	66.7%
Interface interference	0.0%	0.0%	0.0%
Confirm shaming	0.0%	0.0%	0.0%

The effect is concentrated in patterns that already succeed in English. The two patterns that are nulls in English remain nulls in every language—consistent with Section ??’s account, since language cannot make an agent acquire visual attention or a capacity for guilt.

VII. QUALITATIVE ANALYSIS

A. How agents fall for subscription traps

We inspected twenty randomly sampled deceived episodes from SaaS billing, the highest-susceptibility pattern under ComputerUse. All twenty are genuine model failures rather than harness artefacts, in three mechanisms.

Semantic conflation ($\approx 65\%$). The model treats “free trial” as satisfying an instruction to select a “free plan,” not registering that the trial converts to a recurring charge. The auto-renewal terms are present and legible in the extracted context; the model does not connect them to its constraint.

Constraint-violating rationalisation ($\approx 25\%$). At higher intensities no zero-cost option exists. The model identifies this correctly—one trace reads “there’s no clear option for a free plan without auto-renewal”—then clicks the trap anyway, on the grounds it is “the only actionable step.” The constraint was

understood and then discarded under pressure to complete the task.

Keyword-triggered action ($\approx 10\%$). A single-step response matching on “free” without evaluating surrounding pricing.

B. Task abandonment

Excluding the episodes explained in Section V-B, nineteen episodes ended with task incomplete and pattern avoided, in three archetypes.

Principled refusal (11 episodes, SaaS billing, larger model). The model recognises that the only available path commits the user to a recurring charge in violation of its instruction, and declines to act. We regard this as correct behaviour when an interface admits no compliant action, and note it is the larger model exhibiting it.

Premature completion (3 episodes, confirm shaming). The model removes the manipulated add-on, then declares the task finished without completing checkout—resisting the manipulation but losing the task.

Fixation loop (5 episodes, basket sneaking, smaller model). The model identifies a sneaked item and refuses to pay, then repeats the same corrective action for twenty consecutive steps until the budget expires. It detected the constraint and lacked the planning capacity to act on it—a capability boundary, not a susceptibility one.

C. What the four nulls have in common

Four patterns produced no genuine deception at any intensity under either architecture, and we argue this is not coincidence.

Interface interference manipulates visual weight. At aggressive intensity the decline option is low-contrast, de-emphasised text beside a dominant renewal button. Traces show the agent identifying it directly—“the ‘Accept the risk’ button is the option that aligns with this goal”—and selecting it.

Nagging manipulates persistence, interrupting repeatedly for data access. The agent declines and proceeds.

Subscription trap manipulates attrition, routing cancellation through a four-step retention flow. The agent completes all four steps.

Confirm shaming manipulates guilt, labelling the decline option to make choosing it feel callous. The agent quotes the label to locate the button and clicks it.

The four target properties agents do not possess: selective visual attention, irritation, finite patience, and susceptibility to social shame. Every pattern that *does* succeed—auto-renewing trials, negated consent phrasing, late-revealed fees, substituted items—manipulates what information is available or how it is framed, which is the channel through which an agent perceives the game at all.

This marks a division inside the CCPA taxonomy that its legal categories do not: patterns exploiting human perceptual and affective limits, and patterns exploiting information asymmetry. Only the second class transfers. If agent-mediated

commerce grows, the first class loses potency while the second does not—with direct consequences for which categories warrant continued enforcement attention in an agentic setting.

These remain claims about *text-reading* agents. A vision-based agent, for which visual salience is a real signal, could overturn the interface-interference null specifically.

VIII. LIMITATIONS

The identifier audit is incomplete. Section ?? establishes that element identifiers disclosed the answer for two patterns. We have not audited the remaining ten. Any pattern whose identifiers encode a semantic distinction the task depends on is potentially affected, and the reported rates for those patterns would be lower bounds on true susceptibility.

The models under test no longer exist. Both primary agent models were withdrawn by the provider shortly after data collection. The benchmark is reproducible in construction—testbed, specifications, rubrics and configuration enumeration are released—but the figures are not, because the system they measured is gone. Results obtained against hosted proprietary endpoints have a shelf life set by the provider, not the research.

The fully-localised language arms are not interpretable. See Section V-D. We report them for completeness and because the contamination is itself a finding.

BrowserUse outcomes are partly architectural. 11.8% of scored E1b episodes reflect the framework’s completion signal rather than agent behaviour, and five control deceptions remain unexplained. Our cross-architecture claims are restricted to the four nulls, which hold in both.

Two architectures, both DOM-based. Neither is vision-based.

Single provider. The pattern-level ranking may not generalise across model families.

Trace-only judging. The judge scores reasoning traces without the final screenshot, as the selected judge model lacks vision.

The schema did not record a manipulated variable. When the language arms ran, the episode log stored a single language column; the separation of instruction language from interface language existed only in memory and was dropped on insert. The conditions are recoverable only because they ran with disjoint seed blocks on different dates—fortunate rather than designed.

Deterministic outcomes conflate mechanism. For oracle-scored patterns we have no equivalent of the judge’s causal separation, so a deceived score does not distinguish falling for the manipulation from mis-executing a correct intention. This is Section V-D’s failure in a milder form; traces are retained so it can be quantified.

DF is unreachable by construction. *avoided* is carried by the oracle result, so a deceived outcome requires a completed transaction. The four-way outcome space is three-way in practice.

IX. CONCLUSION

We built Armavour to ask whether legally-codified dark patterns work on agents. Some do and some do not, and the

division is not predictable from how well they work on people: patterns exploiting information asymmetry transfer, patterns exploiting human perceptual and affective limits do not.

We also found five ways our own instrument measured something other than what it named. An item whose manipulation was never load-bearing. Element identifiers that told the agent the answer. A scoring rule that read an agent framework’s “task complete” as “dark pattern avoided.” A table whose denominator changed mid-column. And a deception rate inflated by the agent’s inability to read the honest interface. All five produced stable, plausible numbers. Four biased toward apparent safety.

We do not think we were unusually careless. We think these are the shapes measurement failure takes in this class of benchmark, and that they are difficult to see precisely because the numbers they produce look correct. Each was found by the same method: state how a reported quantity was supposed to be computed, recompute it from raw records, and look at where the reconstruction breaks. That method cost us days and found five errors, one of which inflated our own headline result. We suggest it belongs in the standard reporting path for agent safety benchmarks, not in the incident response path.

AVAILABILITY

Testbed, harness, pattern specifications, judge rubrics, validation datasets and analysis code are available at <https://github.com/Soumya-Vinod/Armavour>. Note the model availability caveat in Section ??.

REFERENCES

- [1] Central Consumer Protection Authority, “Guidelines for Prevention and Regulation of Dark Patterns, 2023,” Ministry of Consumer Affairs, Government of India, 2023.
- [2] [SusBench citation — to be completed.]
- [3] [Brignull, dark patterns taxonomy — to be completed.]
- [4] [Mathur et al., dark patterns at scale — to be completed.]
- [5] [Web agent benchmark citation — to be completed.]
- [6] [Multilingual LLM evaluation citation — to be completed.]
- [7] [BrowserUse framework citation — to be completed.]
- [8] [Construct validity in ML evaluation — to be completed.]
- [9] [Benchmark contamination / saturation — to be completed.]